QUOC HUY DINH (RYAN)

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PROFESSIONAL SUMMARY

Artificial Intelligence and Computer Science student at the University of Technology Sydney (UTS) with hands-on experience building AI-powered systems, deep learning models from scratch, Python data pipelines, and full-stack web applications. Currently a Research Assistant working on Vision-Language Models and generative modelling, and leading the ML side of a computer-vision research project targeting international publication. Comfortable taking work end to end, from research and data preprocessing through to shipped products. Open to internship, casual, part-time, contract, and graduate roles across AI, software, and data. Fast learner who delivers strong outcomes both independently and in teams.

EDUCATION

Bachelor of Artificial Intelligence University of Technology Sydney (UTS) | Sydney, Australia | Jul 2025 to Present
- Focus on machine learning, deep learning, and applied AI systems
- Database Fundamentals: High Distinction (100/100), full marks across relational algebra, joins, aggregation, and query optimisation

Bachelor of Computer Science Ho Chi Minh City University of Technology (HCMUT) | Aug 2023 to Jun 2025
- GPA: 3.8 / 4.0
- Data Structures and Algorithms (C++): A grade

EXPERIENCE

Research Assistant University of Technology Sydney | Sydney, NSW | Nov 2025 to Present
- Cleaned and analysed noisy multi-source Optical Character Recognition (OCR) datasets, ran iterative model experiments, and translated findings into reproducible workflows
- Built normalisation, cleaning, and feature-extraction pipelines for downstream modelling across multi-source datasets
- Contributed backend server modules (TCP, OBS WebSocket, and REST API layers) for a game-research project, enabling real-time messaging and programmatic control
- Currently researching Vision-Language Models (VLM) and Generative Adversarial Networks (GAN), running multimodal understanding and generative modelling experiments

AI AND MACHINE LEARNING PROJECTS

Transformer Chatbot from Scratch PyTorch, Lightning, FastAPI, Hugging Face Spaces
- Built a decoder-only transformer entirely from scratch, including a custom tokenizer, causal masking, and temperature sampling, without relying on high-level model libraries
- Architecture: 4 decoder blocks, d_model 256, 4 attention heads, word-level vocabulary trained on a 40-intent dataset
- Served the model through a FastAPI backend and deployed it on Hugging Face Spaces, powering a live chatbot widget on a personal portfolio site

Smart Parking and Intelligent Traffic Management (ALPR) ML Team Lead | PyTorch, Ultralytics YOLO, PARSEQ, ByteTrack, Streamlit
- Leading the machine learning work on a full automatic licence-plate recognition pipeline targeting international publication
- Combined YOLOv11-OBB plate detection, a PARSEQ transformer OCR head, ByteTrack temporal voting, and Zero-DCE++ low-light enhancement
- Benchmarked robustness across CCPD2020 subsets (base, blur, tilt, rain, night, challenge) and built a Streamlit dashboard for real-time violation detection
- Team lead for a group of three across five ablation configurations spanning detector and OCR axes

Set-of-Mark Visual Prompting for Tactical Soccer Understanding Research, target DICTA 2026 | PyTorch, Gemini, Qwen2.5-VL, YOLO, OpenCV
- Designed a training-free recipe that overlays detector marks (numbered player boxes, team colours, ball markers) onto broadcast frames so a Vision-Language Model can reason about possession, open space, passing lanes, and pressing
- Built a Tactical-QA benchmark across six question types and ran ablations against an oracle ceiling derived from SoccerNet game-state reconstruction
- Inference-only on a T4 GPU, using off-the-shelf YOLO and a VLM API, with McNemar and Cohen's kappa as core rigour tools

Dynamic Graph Learning for Esports Win Prediction Research, target top-tier ML venue | PyTorch, PyTorch Geometric
- Designed a real-time win-probability model that represents each minute of a competitive match as a dynamic heterogeneous graph: ten player nodes plus kill and teamfight edges with temporal decay
- Used a Graph Attention Network plus GRU architecture to output a per-minute win-probability curve and analyse turning-point moments

Machine Learning Models from Scratch Python, NumPy
- Implemented regression and classification algorithms from first principles, applying gradient descent and regularisation, and analysed the bias-variance trade-off and convergence behaviour

SOFTWARE ENGINEERING PROJECTS

Personal Portfolio Site (this website) Next.js 16, TypeScript, Tailwind CSS, Framer Motion, Resend, FastAPI
- Built a multi-page Next.js portfolio (App Router) with shared layout for navbar, footer, and a chatbot widget, plus dedicated routes for resume, projects, research, and contact
- Integrated a Resend-powered contact form with server-side validation, and proxied chatbot requests through a Next.js API route to a separate FastAPI service deployed on Hugging Face Spaces
- Implemented dark mode by default, responsive layout, route-based active navigation, and project filtering by industry
- Live in production at cv-website-lemon.vercel.app, deployed on Vercel with CI on every push to main

AI-Powered Coffee Shop Manager Dashboard Python, LangChain, Redis, Docker, AWS (planned)
- Designed and built an end-to-end AI briefing system delivering automated morning reports covering revenue trends and operational insights, removing the need for manual data review
- Developed a single LangChain pipeline integrating four AI capabilities: natural language Q&A, revenue summarisation, inventory forecasting, and scheduled auto-generated daily reports

- Implemented a Redis caching layer to serve repeated queries without redundant recomputation, and containerised the system with Docker for consistent deployment

CurricuLLM, Parent-Teacher AI Platform Cambridge EduX Hackathon 2026, Team of 4 | TypeScript, React, FastAPI, PostgreSQL

- Built an AI pipeline that auto-generates personalised curriculum progress reports for parents from teacher input, removing manual report writing
- Developed a parent-facing AI chat interface for natural language queries on student progress, served via FastAPI, with real-time multilingual translation
- Contributed full-stack across the React frontend and the FastAPI/PostgreSQL backend (REST API design, database schema)

Game Automation and Streaming Integration Tool Python, Flask, FastAPI, TCP, OBS WebSocket, REST API

- Implemented a TCP layer for real-time client-server communication across two repositories (main app and standalone TCP server/client), enabling low-latency bidirectional messaging
- Integrated the OBS WebSocket protocol to react to live stream events and exposed keyboard/mouse triggers via REST API endpoints across a modular backend
- Designed the boundary between control plane and event plane so the OBS integration and the keyboard/mouse triggers can be extended independently

E-Commerce Web Application React, Express.js, Node.js, PostgreSQL, JWT

- Built a full-stack e-commerce platform independently, covering product listing, search and filter, shopping cart, and checkout with order management
- Implemented JWT-based authentication (register, login, protected routes) with bcrypt password hashing
- Designed and normalised a relational PostgreSQL schema (up to 3NF) and optimised queries for CRUD operations

Frontend Projects React, JavaScript, HTML, CSS

- Built a React Tic-Tac-Toe game with move history and undo (deployed on Netlify), a React Music Player, and a React Weather App
- Built pure HTML/CSS/JS UI clones (Steam-style marketplace and job-search interfaces) focused on responsive layout and accessible markup

DATA AND ANALYTICS PROJECTS

Game Telemetry Data Analysis Python, Pandas, NumPy, Matplotlib, Seaborn

- Ran exploratory data analysis on competitive-game telemetry: cleaning, feature engineering, and statistical analysis of player behaviour and match outcomes, surfacing actionable performance and outlier insights

OCR Preprocessing Pipeline Python, OpenCV, NumPy, PaddleOCR

- Built an image preprocessing pipeline (adaptive thresholding, denoising, deskewing, contrast enhancement) and benchmarked downstream OCR accuracy across configurations to find the strongest pipeline for noisy inputs

Housing Price Data Analysis Python, Pandas, scikit-learn, Matplotlib

- Performed EDA, missing-value handling, and statistical modelling to identify the strongest predictors of house price, using hypothesis testing, correlation analysis, and regression diagnostics

SKILLS

Programming Languages: Python, C++, JavaScript, TypeScript, HTML/CSS
AI / Machine Learning: PyTorch, PyTorch Geometric, Lightning, scikit-learn, LangChain, Ultralytics YOLO, Natural Language Processing (NLP), Large Language Models (LLM), Vision-Language Models (VLM), GANs, Transformers, Data Cleaning, Feature Extraction
Frameworks and Libraries: Next.js, React, Tailwind CSS, Flask, FastAPI, Express.js, Node.js, Pandas, NumPy, OpenCV, Streamlit, Matplotlib, Seaborn
Databases: PostgreSQL, Redis
Tools and Platforms: Vercel, Docker, Git, Jupyter, VS Code, Hugging Face Spaces, AWS (planned)
Protocols and Integration: TCP/IP, OBS WebSocket, REST APIs, JWT Authentication, Resend